

Driftcontrol in AthenaK

This document collects the driftcontrol algorithms used in AthenaK to reduce the black hole kick in various applications.

Driftcontrol

PRELIMINARIES In black-hole neutron star mergers the remnant black hole experiences a physical kick during merger which leads to it moving away from the coordinate origin. For analysis purposes this is not ideal, since the matter distribution will follow the compact object and in most cases the surface analysis for ejecta etc. is biased. In short simulations this might not make a tremendous impact, but it certainly does in longer evolutions. As such we can seek to modify the gauge to pull the black hole back to the desired target location and keep it there throughout the rest of the simulation. Due to the gauge-invariance of the system this should not alter any physical properties. AthenaK implements four different methods that address this problem and which are described below. Here, we first introduce the basic concept behind the driftcontrol.

BASIC CONCEPT Consider that the drifting remnant black hole is followed by AthenaK's `CompactObjectTracker`, so we now it's position and velocity over time. For a tracked black hole the velocity of the tracker is determined as $v_{\text{tracker}}^i = -\beta^i$ and the position is computed via $x_{\text{tracker}}^i = x_{\text{tracker,old}}^i + \Delta t v_{\text{tracker}}^i$. AthenaK's driftcontrol never touches the position, but instead adds a controlling term to the RHS of the shift evolution equation β^i . In particular, we seed something like

$$\partial_t \beta^i = \dots - f_{\text{corr}} g, \quad (1)$$

where g is a Gaussian window, and $f_{\text{corr}} = -u$ where u is what each individual controller seeds based on the specified algorithm. For a unit-mass particle we find

$$\dot{x}^i = -\beta^i, \quad (2)$$

$$\ddot{x}^i = -\partial_t \beta^i = -u + f, \quad (3)$$

where f is the unperturbed RHS of the shift evolution equation, i.e. the contribution from the Γ -driver, shift advection, and η -damping. We therefore inject a control acceleration u into the system that tries to control the drifting black hole. The algorithm is as follows:

1. Measure the error of the current position to the target position $e = x - x_{\text{target}}$.
2. Compute the injected acceleration $u(e, v, \text{state})$ defined by each control mechanism.
3. Multiply that by a Gaussian window $g(r)$ which ideally is a certain factor larger than the black hole.
4. Feed this into the evolution equation of the shift $\partial_t \beta + = u \cdot g$.
5. The velocity will be automatically adjusted via $\dot{x}^i = -\beta^i$.
6. Steps 1-5 are repeated at every cycle at the end of every RK stage.

Physically nothing changes, as the shift is pure gauge and does not enter the constraints. However, holding the black hole there introduces additional coordinate shear around the black hole because the coordinates must keep stretching to keep the object there.

SHARED PROPERTIES The four driftcontrol algorithms that will be introduced below all share some common machinery inside of AthenaK:

- `dc_tracker_index` selects which object shall be controlled (after merger both share the same object).
- `dc_fixed_x/y/z` is the target position we want the black hole at.
- `dc_vel` is the velocity computed as the finite-difference of the tracker position between cycles, i.e. $v = \frac{x_n - x_{n-1}}{\Delta t}$, but it is not the tracker's own velocity. This velocity is clamped to $\pm \text{dc_vel_cap}$, which guards against bad interpolations or a tracker jump from injecting a huge derivative kick into the shift.
- The correction f_{corr} we feed into the shift is spatially constant and only the weight varies. With the Gaussian window we put the shear, where the peak of the Gaussian window is located (ideally at the black hole). `dc_gaussian_center` centers the Gaussian window on either `fixed` (the target position) or `tracker` (the tracked object). `dc_damping_scale` is the $\frac{1}{e}$ radius inside of $g(r) = \exp\left(\frac{-r^2}{s^2}\right)$, where s is the damping scale.

Oscillator

FORMULATION The first method that is implemented is the Oscillator method from THC (second-order). For this method, the injected acceleration into the shift evolution is given by

$$u = \frac{e}{\tau^2} + \frac{2\zeta v}{\tau}, \quad (4)$$

which is a natural damped harmonic oscillator with frequency $\omega = \frac{1}{\tau}$, and where e is the error of the current position with respect to the target. It is unconditionally stable for $\zeta, \tau > 0$. The parameters that can be tuned are

- **dc_damping_time**: the restoring timescale $\tau = \frac{1}{\omega}$. A small τ means a stiff, and fast spring. This can lead to pullback speeds exceeding the speed-of-light. At critical damping the compact object settles in roughly $3-5\tau$.
- **dc_damping_coeff**: the damping ratio ζ . $\zeta < 1$ represents an underdamped oscillator, $\zeta = 1$ represents critical damping, and $\zeta > 1$ overdamping. The default is critical damping.

CAVEAT Equ. (4) can be written as

$$\ddot{x} + \frac{e}{\tau^2} + \frac{2\zeta}{\tau}\dot{x} = f. \quad (5)$$

Setting $\ddot{x} = v = 0$ gives $e = f\tau^2$, which means with this method we retain a constant offset that scales with the square of the timescale. Halving the offset costs a factor of $\sqrt{2}$ in stiffness. One can use it when the drift is mild and a constant offset is acceptable. It works better if it is turned off shortly after the kick, such that the initial separation from the target is not too large yet.

PID

FORMULATION The PID method is third-order and the injected acceleration is defined by

$$u = K_p e + K_i \int e dt + K_d v, \quad (6)$$

where u without the integral term and $K_p = \frac{1}{\tau^2}$, $K_d = \frac{2\zeta}{\tau}$ reproduce the harmonic oscillator of Equ. (4) exactly. The integral term accumulates offset from the target and keeps pushing until the offset is gone, which kills the problem the Oscillator method had. This extended method requires that the coefficients $K_i, K_p, K_d > 0$ and $K_i < K_p K_d$. The individual parameters that can be tuned are:

- **dc_Kp**: the spring constant $\omega_0 = \sqrt{K_p}$, with the default being 1.0 such that $\tau = 1$.
- **dc_Kd**: the damping parameter $\zeta = \frac{K_d}{(2\sqrt{K_p})}$ (default is critically damped).
- **dc_Ki**: the integral gain, i.e. how fast the accumulated offset is converted into a restoring force.
- **dc_integral_cap**: clamp on the integral $\int e dt$.

CAVEAT The residual error in principle is now zero with this method. However, releasing the driftcontrol at one point is fatal, because the restoring force used to keep the object at the desired location is then used to kick the black hole away again with the opposite force. The only option is to keep the remnant at the location with PID on for the entire time.

Relaxation

FORMULATION The Relaxation method is first-order and relaxes the local shift field toward a commanded value. It modifies the shift evolution equation in the form

$$\partial_t \beta^i = \left(\frac{1}{\tau_r} \right) (\beta^i - \kappa e^i) g - G_i \Gamma^i g, \quad (7)$$

where g again is the Gaussian window, τ_r is the timescale, and G_i is a constant. If $\frac{1}{\tau_r}$ dominates the other gauge terms, then $\beta^i \rightarrow \kappa e^i$ on timescale τ_r . Then $\dot{x} = -\beta = -\kappa e$ and the offset decays exponentially at rate κ . The tuning parameters are:

- `dc_kappa`: return rate κ to the desired location.
- `dc_relaxation_time`: relaxation time τ_r similar to the Oscillator method.
- `dc_gamma_suppress`: fades out the Γ -driver's contribution to $\partial_t \beta^i$ inside the window (no suppression at $e = 0$). In essence, the relaxation term and the Γ -driver are fighting and in principle, the further off target one is the more the driftcontrol should win.

■ **CAVEAT** At equilibrium $\beta = 0$ we get a residual $e = \frac{f\tau_r}{\kappa}$, which is still biased but still much better than the Oscillator at the same τ_r . This is the only of the four methods that locally changes the shift gauge instead of nudging it. It is also not guarded against superluminal coordinate speeds. However, it pins the object locally.

DOB (Disturbance observer)

■ **FORMULATION** The DOB method is third-order and it uses the injected acceleration

$$\hat{f} = p + \omega_0 v, \quad (8)$$

$$u = \omega_c^2 e + 2\zeta\omega_c v + \hat{f}, \quad (9)$$

$$\dot{p} = \omega_0(\omega_c^2 e + 2\zeta\omega_c v). \quad (10)$$

We can re-arrange the equation for u to $\omega_c^2 e + 2\zeta\omega_c v = u - \hat{f}$, which leads to the identity $\dot{p} = \omega_0(u - \hat{f})$. Differentiating $\hat{f}$ leads to $\dot{\hat{f}} = \dot{p} + \omega_0 \hat{f}$. Substituting $\dot{p}$ and the controller $\ddot{x} = -u + f$ results in $\dot{\hat{f}} = \omega_0(f - \hat{f})$, i.e. the estimator does not depend on what the controller is doing. The first two terms in u are essentially just what the Oscillator method is doing. To understand the role of $\hat{f}$ a bit better, we can split the injected acceleration in two parts

$$u = u_{\text{PD}} + \hat{f}, \text{ PD=Proportional-Derivative} \quad (11)$$

$$u_{\text{PD}} = \underbrace{\omega_c^2 e}_{\text{proportional to } e} + \underbrace{2\zeta\omega_c v}_{\text{derivative of } e}, \quad (12)$$

which if substituted into $\ddot{x} = -u + f$ gives $\ddot{x} = -u_{\text{PD}} + (f - \hat{f})$. $\hat{f}$ therefore is trying to delete f from the equation. Physically, the observer is inferring the drive from the mismatch between the acceleration that wanted and the acceleration one actually gets. Put differently, we push with $-u$, watch the puncture and then attribute the difference to the ambient gauge, i.e. f . The tuning parameters are:

- `dc_omega_c`: the analog to $\frac{1}{\tau}$ given in units $[\text{M}^{-1}]$.
- `dc_zeta`: the damping ratio, and identical to the Oscillator method. Critical damping, i.e. $\zeta = 1.0$ means there is no overshoot.
- `dc_omega_o`: observer bandwidth, i.e. how fast $\hat{f}$ tracks f . This improves the disturbance rejection without eating any stability margin. It is numerically limited by $\omega_0 \Delta t < 0.5$.

■ **CAVEAT** It is the preferred method over PID and the others, because it has no ceiling for instance on the integral as PID and if the state is ever lost or wrong, it re-converges in $\approx \frac{1}{\omega_0}$.